

Laser microdissection (LMD) can aid in the diagnosis of diseases associated with heterogenous cell types or abnormal subcellular structures. LMD enables genetic analysis of specific cells without interference from healthy tissues. Tissue samples are examined under a microscope to identify suspicious regions, which are separated from the sample using lasers.

There are two common LMD methods for separating suspicious cells. One method uses an ultraviolet laser with a wavelength of 300 nm to ablate all the unwanted tissue, leaving only the cells of interest for further analysis.

The other method uses an infrared laser with a wavelength of 800 nm and a thermoplastic film, as shown in Figure 1. The thermoplastic film is placed over the sample and the suspicious regions are exposed to the laser, which heats the thermoplastic so it sticks to the underlying tissue. Removing the thermoplastic separates the cells of interest from the sample.

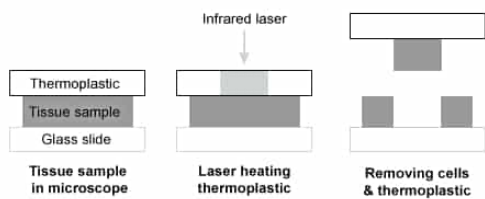


Figure 1 LMD system with infrared laser and thermoplastic for separating cells from a tissue sample

Researchers measured the temperature change of different thermoplastic samples exposed to an infrared laser; the findings are shown in Table 1.

Table 1 Heating of Different Thermoplastics with an Infrared Laser

Thermoplastic	Mass (g)	Heat (J)	Temperature change (°C)
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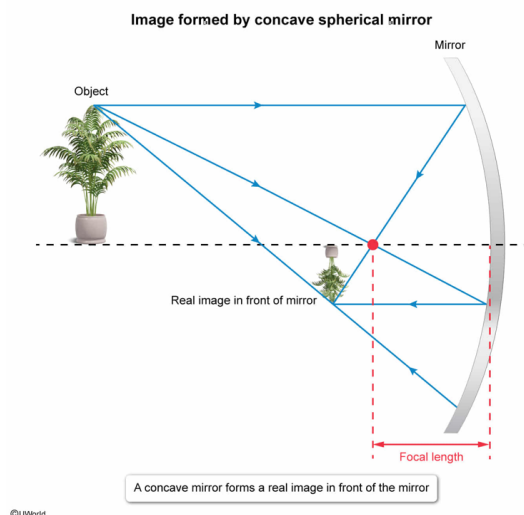
The optical cavity of a laser is 6 cm long. What kind of image is formed by one of the mirrors from this cavity when the object is outside the focal length?

- ☐ A. Virtual image behind the mirror [20%]
- ☐ B. Real image behind the mirror [13%]
- ☐ C. Virtual image in front of the mirror [18%]
- ☒ D. Real image in front of the mirror [47%]

Correct

46% answered correctly

Explanation:



Spherical mirrors are shaped like the surface of a sphere. For a **concave mirror** (ie, a converging mirror), the reflective surface curves inward. In contrast, for a **convex mirror** (ie, a diverging mirror), the reflective surface curves outward.

When the distance between the object and mirror is greater than the focal length, a concave mirror reflects the light rays from the object so they converge in front of the mirror and form a **real image**. However, a convex mirror reflects the light rays so they diverge and form a **virtual image** behind the mirror.

The passage states that the optical cavity of a laser contains two concave spherical mirrors. Therefore, the mirror forms a real image in front of the mirror.

(Choices A and C) Virtual images are formed by convex mirrors, but the optical cavity of a laser uses concave mirrors.

(Choice B) Real images are formed in front of the mirror, not behind the mirror.

Educational objective:

A concave (converging) mirror creates a real image of an object positioned outside the mirror's focal length, whereas a convex (diverging) mirror creates a virtual image. Real images form in front of the mirror, and virtual images form behind the mirror.

Time spent: 0 sec
Subject:
Physics

QID: 403844
System:
Light and Sound

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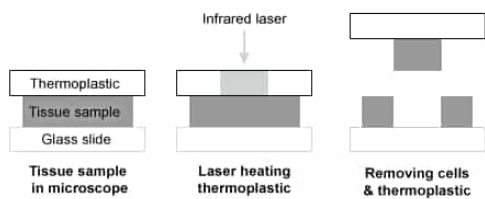


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According to the data in Table 1, what is the specific heat of the Sample 2 thermoplastic?

- ☐ A. 2000 J/(kg·°C) [7%]
- ☐ B. 1200 J/(kg·°C) [2%]
- ✔ ☒ C. 1000 J/(kg·°C) [82%]
- ☐ D. 400 J/(kg·°C) [6%]

Correct

82% answered correctly

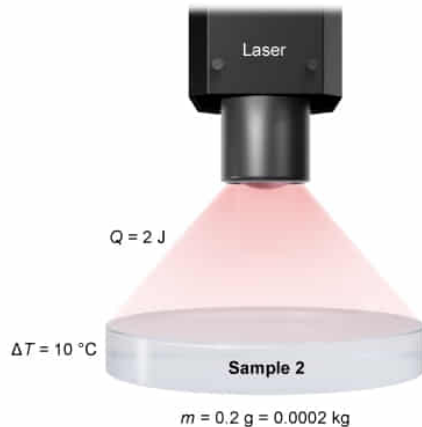
Explanation:

Specific heat of Sample 2 thermoplastic

$$Q = mc \cdot \Delta T$$

$$c = \frac{Q}{m \cdot \Delta T}$$

- Q Heat
- m Mass
- c Specific heat
- ΔT Change in temperature



Substitute Q , m , & ΔT into equation above to calculate c .

$$c = 1000 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}$$

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According to the **calorimetry** equation, the amount of **heat** Q transferred into an object equals the product of the object's **mass** m , **specific heat** c , and change in **temperature** ΔT :

$$Q = mc \Delta T$$

The calorimetry equation can be rearranged to solve for c , yielding:

$$c = \frac{Q}{m \Delta T}$$

In the passage, [Table 1](#) lists the following values for the Sample 2 thermoplastic: $m = 0.2 \text{ g}$, $Q = 2 \text{ J}$, and $\Delta T = 10^\circ\text{C}$. Substituting these values into the equation above yields:

$$c = \frac{2 \text{ J}}{(0.2 \text{ g})(10^\circ\text{C})} = 1 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$

Converting this value into the units of $\text{J}/(\text{kg} \cdot ^\circ\text{C})$ gives:

$$c = \left(1 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}\right) \left(1000 \frac{\text{g}}{\text{kg}}\right) = 1000 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}$$

Therefore, the Sample 2 thermoplastic has a c of $1000 \text{ J}/(\text{kg} \cdot ^\circ\text{C})$.

(Choice A) $2000 \text{ J}/(\text{kg} \cdot ^\circ\text{C})$ is the specific heat of the Sample 1 thermoplastic, not the Sample 2 thermoplastic.

(Choice B) $1200 \text{ J}/(\text{kg} \cdot ^\circ\text{C})$ is the specific heat of the Sample 3 thermoplastic, not the Sample 2 thermoplastic.

(Choice D) $400 \text{ J}/(\text{kg} \cdot ^\circ\text{C})$ is calculated by using 0.5 g for the mass instead of 0.2 g .

Educational objective:

The calorimetry equation calculates the heat energy required to raise the temperature of an object based on the object's mass, specific heat, and change in temperature. An object's specific heat equals the ratio of the heat and the product of the mass and change in temperature.

Time spent:0 sec**Subject:**
Physics**QID:**403845**System:**
Light and Sound

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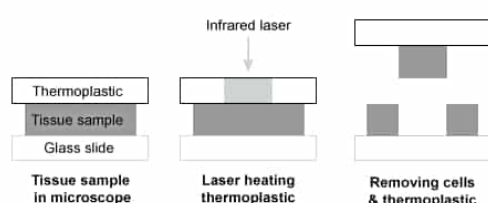


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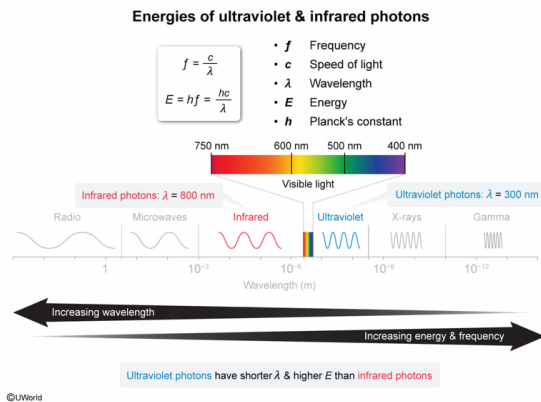
Which of the following statements best explains why an ultraviolet laser is used to ablate unwanted cells from the sample instead of an infrared laser?

- ☐ A. The photon wavelength is longer for the ultraviolet laser than the infrared laser. [4%]
- ✓ ☒ B. The photon energy is higher for the ultraviolet laser than the infrared laser. [88%]
- ☐ C. The photon frequency is lower for the ultraviolet laser than the infrared laser. [5%]
- ☐ D. The photon velocity is lower for the ultraviolet laser than the infrared laser. [1%]

Correct

87% answered correctly

Explanation:



Lasers emit photons of **electromagnetic radiation** over a very narrow range of **wavelengths** λ . The **frequency** f of the photons equals the ratio of the speed of light c and λ :

$$f = \frac{c}{\lambda}$$

Furthermore, the **energy** E of the laser photons equals the product of Planck's constant h and f :

$$E = hf = \frac{hc}{\lambda}$$

Hence, E is **inversely proportional** to λ .

The passage states that the λ of the ultraviolet laser is 300 nm and the λ of the infrared laser is 800 nm. Therefore, E is higher for the ultraviolet laser than the infrared laser because it has a shorter λ .

(Choice A) The photon wavelength of the ultraviolet laser is shorter than that of the infrared laser, not longer.

(Choice C) The photon frequency of the ultraviolet laser is higher than that of the infrared laser, not lower.

(Choice D) The photon velocity of the ultraviolet laser is identical to that of the infrared laser, not lower.

Educational objective:

Lasers emit photons of electromagnetic radiation with a narrow range of wavelengths. Photons with a shorter wavelength have higher frequency and energy.

Time spent: 0 sec

Subject:

Physics

QID: 403846

System:

Light and Sound

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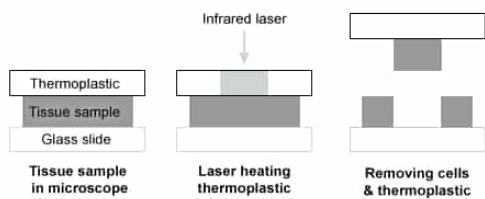


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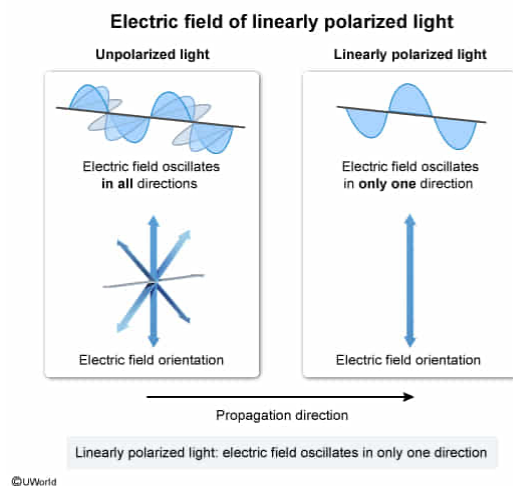
Linearly polarized light can be used to improve the contrast of certain tissue samples viewed through a microscope. The advantage of linearly polarized light over nonpolarized light is that:

- ☐ A. all electric field oscillations are eliminated. [3%]
- ☒ B. all electric field oscillations are aligned along one specific direction. [88%]
- ☐ C. the amplitude of the electric field oscillations is increased. [5%]
- ☐ D. the frequency of the electric field oscillations is increased. [2%]

Correct

88% answered correctly

Explanation:



Visible light is a form of electromagnetic radiation consisting of oscillating **electric and magnetic fields** that are perpendicular to each other and perpendicular to the direction of propagation. Typical sunlight or light from an electric bulb is **unpolarized**, so the electric field oscillates in all directions.

Light that is **linearly polarized** refers to the special case where the electric field oscillates only in one direction. Consider an xyz-coordinate system where the light propagates along the x-axis. For **linearly polarized light**, the electric field may oscillate only along the y-axis or only along the z-axis. Consequently, a **linear polarization filter** allows only the transmission of light oriented in **one direction** and blocks all light with other orientations.

In this question, a microscope can use linearly polarized light to improve the contrast of tissue samples. The distinguishing feature of linearly polarized light compared to unpolarized light is that the electric field oscillations are aligned along one specific direction.

(Choice A) Eliminating all electric field oscillations would also eliminate all the light.

(Choice C) Polarization reduces the amount of light and decreases (not increases) the amplitude of the electric field oscillations.

(Choice D) Polarization does not change the frequency of the electric field oscillations, only the direction of the oscillations.

Educational objective:

Unpolarized light consists of an electric field that oscillates in all directions that are perpendicular to the direction of propagation. In contrast, the electric field of linearly polarized light oscillates in only one direction.

Time spent: 0 sec

Subject:
Physics

QID: 403847

System:
Light and Sound

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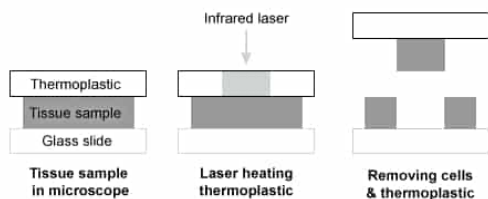


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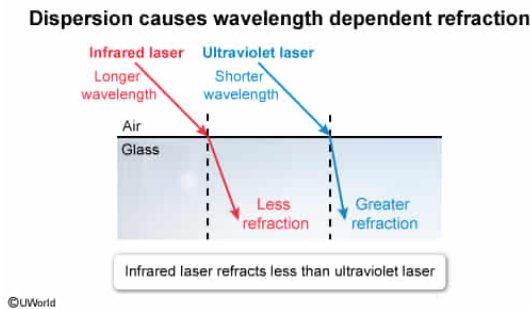
Which of the following describes the refraction of infrared and ultraviolet lasers when they travel from air into a glass slide holding a sample?

- ☐ A. The refraction is greater for the infrared laser than the ultraviolet laser. [26%]
- ☐ B. The refraction for the two lasers is equal but not zero. [31%]
- ☒ C. The refraction is less for the infrared laser than the ultraviolet laser. [38%]
- ☐ D. The refraction for the two lasers is zero. [2%]

Correct

38% answered correctly

Explanation:



Refraction causes light rays to bend when they travel from one medium into another medium with a different **index of refraction**. [Snell's Law](#) describes the relationship between the angle of incidence θ_1 , the angle of refraction θ_2 , the index of refraction for the original medium n_1 , and the index of refraction for the new medium n_2 :

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

However, the values of n_1 and n_2 are slightly different for different **wavelengths** of **electromagnetic radiation**. As a result, longer wavelengths refract less than shorter wavelengths. This phenomenon, **dispersion**, causes white light passing through a glass prism to separate into its constituent colors.

In this question, the infrared laser has a longer wavelength than the ultraviolet laser. Therefore, the infrared laser refracts less than the ultraviolet laser at the interface between the air and glass.

(Choice A) The infrared laser does not refract more than the ultraviolet laser because it has a longer wavelength compared to the ultraviolet laser.

(Choice B) The refraction of the two lasers is not equal because they have different wavelengths.

(Choice D) The refraction of the two lasers is not zero because all electromagnetic radiation refracts when crossing a boundary between two materials with different refractive indices.

Educational objective:

Dispersion is caused because different wavelengths of electromagnetic radiation have slightly different refractive indices. A laser with a longer wavelength refracts less when passing from air into glass.

Time spent: 0 sec
Subject:
Physics

QID: 403848
System:
Light and Sound

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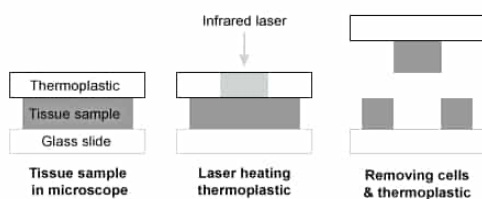


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A researcher uses a microscope with a convex lens to view a sample. The distance from the sample to the lens is 5 cm, and the distance from the lens to the retina is 20 cm. If the distance from the sample to the lens increases by 15 cm, the image is now focused:

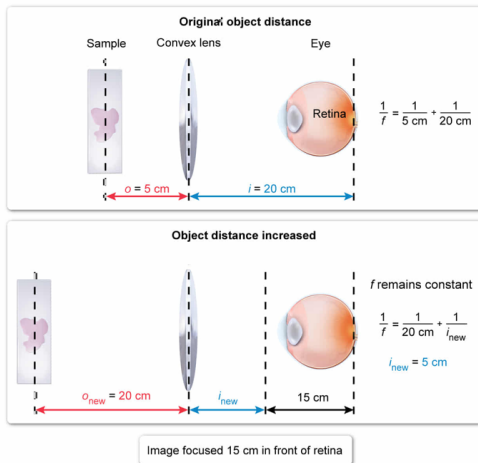
- ✔ ☒ A. 15 cm in front of the retina. [29%]
- ☐ B. 5 cm in front of the retina. [34%]
- ☐ C. 5 cm behind the retina. [22%]
- ☐ D. 15 cm behind the retina. [13%]

Explanation:

Image distance when object distance increases

$$\frac{1}{f} = \frac{1}{o} + \frac{1}{i}$$

- f Focal length
- o Object distance
- i Image distance



A **convex lens**, also called a **converging lens**, has both surfaces curved outward. The **focal length** f of a convex lens is related to the distance from the object to the lens o and the distance from the lens to the image i by the **thin lens equation**:

$$\frac{1}{f} = \frac{1}{o} + \frac{1}{i}$$

In this question, o equals 5 cm and i equals 20 cm, giving:

$$\frac{1}{f} = \frac{1}{5 \text{ cm}} + \frac{1}{20 \text{ cm}}$$

The value of o increases by 15 cm, from 5 cm to 20 cm, creating a new object distance o_{new} and a new image distance i_{new} :

$$\frac{1}{f} = \frac{1}{o_{\text{new}}} + \frac{1}{i_{\text{new}}}$$

$$\frac{1}{f} = \frac{1}{20 \text{ cm}} + \frac{1}{i_{\text{new}}}$$

The two equations above are both equal to the inverse of f , so the right sides of the equations are also equal:

$$\frac{1}{20 \text{ cm}} + \frac{1}{i_{\text{new}}} = \frac{1}{5 \text{ cm}} + \frac{1}{20 \text{ cm}}$$

Subtracting the term $\frac{1}{20 \text{ cm}}$ from each side yields:

$$\frac{1}{i_{\text{new}}} = \frac{1}{5 \text{ cm}}$$

Therefore, i_{new} equals 5 cm. The distance from the convex lens to the retina still equals 20 cm. Hence, the image is focused 15 cm in front of the retina.

(Choices B and C) The image is focused 5 cm from the convex lens, not 5 cm in front of or behind the retina.

(Choice D) The image is focused 15 cm in front of the retina, not behind the retina.

Educational objective:

The focal length of a convex lens is related to the distances from the lens to the object and the lens to the image by the thin lens equation. Increasing the distance to the object decreases the distance to the image.

Time spent:0 sec

Subject:
Physics

QID:403849

System:
Light and Sound